

BAHRIA UNIVERSITY
Islamabad Campus (H-11)

DBMS Lab — Project Proposal

**Hospital Pharmacy &
Medicine Inventory
Management System**

Course	DBMS Lab
Instructor	Engr. Mahrukh Shakoor
Semester / Section	4th Semester — Section B
Submission Deadline	12-05-2026
Group Member 1	Muhammad Hussain — 02-131242-107
Group Member 2	Ali Ahmed Mansoor — 01-131222-009

1. Introduction

Healthcare institutions in Pakistan and across the developing world continue to face severe challenges in managing pharmaceutical inventories. Hospitals and clinics routinely suffer losses due to medicine expiration, stock mismanagement, and the absence of real-time visibility into supply levels. Manual registers and spreadsheets remain the norm in many facilities, leading to dispensing errors, delayed procurement, and inadequate patient care.

This project proposes the design and development of a Hospital Pharmacy & Medicine Inventory Management System — a relational database-driven application that addresses these real-world problems through structured data management, automated alerts, and efficient query-based reporting.

The system will be built as a full-stack **web application** integrating a React.js frontend with a Node.js/Express.js backend and a MySQL relational database. The project strictly adheres to the DBMS Lab requirements, demonstrating normalization, relational integrity, query optimization, and seamless frontend-to-backend data integration.

2. Problem Statement

The following critical problems exist in pharmacy management at most hospitals and clinics today:

- Medicines expire on shelves because expiry dates are not tracked in real time, resulting in financial loss and potential patient harm.
- Stock levels are unknown until a medicine runs out entirely, disrupting patient care at critical moments.
- Purchase orders are placed manually, often without accurate knowledge of current stock, leading to over-ordering or under-ordering.
- Dispense records are maintained on paper, making it impossible to audit which patient received which medicine and at what dosage.
- There is no centralized system linking prescriptions, patient records, doctor information, and dispensing activities.

This system directly solves all five problems through a structured relational database with automated expiry tracking, low-stock alerts, purchase order management, and a complete dispense audit trail.

3. Project Objectives

The primary objectives of this project are:

1. Design a fully normalized relational database (3NF) with 13 interrelated tables covering all aspects of pharmacy operations.
2. Implement complete CRUD operations using SQL with proper constraints, foreign keys, and indexes.
3. Develop a React.js frontend providing an intuitive interface for all system features including dashboard, medicine management, supplier management, and expiry alerts.
4. Build a Node.js/Express.js REST API backend that serves as the bridge between the frontend and the MySQL database.
5. Write optimized SQL queries utilizing JOINS across multiple tables, demonstrating relational database proficiency.

6. Generate meaningful reports including monthly dispense reports, low-stock summaries, and expiry alert summaries.
7. Export the complete database as a portable .sql file that can be imported into any MySQL or compatible RDBMS instance.

4. Project Scope

4.1 In Scope

- Medicine management: add, update, delete, and search medicines with full attribute tracking.
- Inventory/stock management with real-time quantity updates on every dispense operation.
- Supplier management and purchase order lifecycle (Pending → Delivered → Cancelled).
- Patient and doctor registration linked to prescription management.
- Medicine dispensing with automatic stock deduction and full audit trail.
- Expiry alert system categorized by urgency: Expired, Expiring in 7 days, Expiring in 30 days.
- Report generation for dispense history, low-stock items, and supplier order history.
- System settings for pharmacy profile, stock thresholds, and user preferences.

4.2 Out of Scope

- Billing, invoicing, or financial accounting modules.
- Integration with external hospital management systems.
- Mobile application development.
- Multi-branch or multi-hospital deployments.

5. System Features

#	Module	Description
1	Dashboard	Live stats: total medicines, low stock count, expired medicines, pending orders. Bar chart of top medicines by stock. Recent dispense table and expiry alert panel.
2	Medicine Management	Add, edit, delete, and search medicines. Category, manufacturer, dosage form, unit price, expiry date, batch number. Status badges: In Stock / Low Stock / Expired.
3	Supplier Management	Manage supplier profiles including company name, contact person, phone, email, and city. Active/Inactive status toggle.
4	Purchase Orders	Create purchase orders linked to suppliers and medicines. Track order status: Pending, Delivered, Cancelled. Mark as delivered to update stock automatically.
5	Prescriptions & Dispense	Record prescriptions from doctors for patients. Dispense medicines against prescriptions with automatic stock deduction and complete dispense history.
6	Expiry Alerts	Dedicated alert page grouping medicines by: Already Expired, Expiring within 7 days, and Expiring within 30 days. Shows supplier contact for urgent re-ordering.

7	Reports	Generate and export: Monthly Dispense Report, Low Stock Report, Supplier Order History, and Expiry Summary Report. CSV and print export support.
8	Settings	Pharmacy profile, stock threshold configuration, user account management, notification preferences, and system preferences including dark mode and currency.

6. Database Design Overview

The system will use a fully normalized relational database comprising 13 tables. The schema is designed to 3rd Normal Form (3NF) to eliminate data redundancy while maintaining referential integrity through foreign key constraints.

6.1 Database Tables

Table Name	Purpose
categories	Stores medicine categories (Tablet, Syrup, Injection, Capsule, etc.)
manufacturers	Medicine manufacturing company details
suppliers	Supplier company and contact information
medicines	Core medicine details: name, price, expiry, dosage form, batch number
stock	Current stock quantity and low-stock threshold per medicine
stock_transactions	Full audit log of every stock IN and OUT movement
purchase_orders	Purchase order header: supplier, dates, and status
purchase_order_items	Line items within each purchase order
patients	Patient registration and contact details
doctors	Doctor profiles and specializations
prescriptions	Prescription records linking patients and doctors
prescription_items	Individual medicines within each prescription
dispense_records	Complete record of every medicine dispensed

6.2 Key Relationships

- medicines → categories, manufacturers, suppliers (Many-to-One)
- medicines → stock (One-to-One)
- medicines → stock_transactions, dispense_records (One-to-Many)
- suppliers → purchase_orders (One-to-Many)
- purchase_orders → purchase_order_items (One-to-Many)
- patients → prescriptions (One-to-Many)
- doctors → prescriptions (One-to-Many)
- prescriptions → prescription_items, dispense_records (One-to-Many)

7. Technology Stack

Layer	Technology	Purpose
Database	MySQL Community Server	Primary RDBMS — free, industry-standard
DB Design Tool	MySQL Workbench	Schema design, query execution, ER diagram export
ER Diagram	dbdiagram.io	Database schema visualization using DBML code
UML Diagrams	Mermaid.live / draw.io	Use case, class, and architecture diagrams
Frontend	React.js + Tailwind CSS	User interface — all 8 system pages
Backend	Node.js + Express.js	REST API connecting frontend to MySQL
DB Connector	mysql2 (npm package)	Executes SQL queries from Node.js
Code Editor	Google Antigravity / VS Code	Development environment
API Testing	Thunder Client	Testing REST API endpoints
Version Control	Git + GitHub	Source code management

8. Project Work Plan

Week	Phase	Tasks	Deliverable
Week 1	Planning & Design	ER Diagram, Use Case Diagram, Class Diagram, System Architecture Diagram	<i>All UML diagrams complete</i>
Week 2	Database Development	CREATE TABLE statements, constraints, indexes, normalization, sample data insertion	<i>Working MySQL database</i>
Week 3	Frontend Development	React.js UI — all 8 pages functional, Tailwind CSS styling, component architecture	<i>Running frontend on localhost</i>
Week 4	Backend & Integration	Node.js/Express REST API, mysql2 connection, API endpoints, frontend-DB integration	<i>Full-stack system working</i>
Week 5	Testing & Report Writing	End-to-end testing, bug fixes, 5-page project report, database .sql export	<i>Report + exported DB file</i>
Week 6	Viva Preparation	Practice demo, query walkthroughs, mock viva between group members	<i>Viva-ready system & demo</i>

9. Work Division

Responsibility	Muhammad Hussain 02-131242-107	Ali Ahmed Mansoor 01-131222-009
Primary Role	Database Engineer	Full-Stack Developer
Database Schema	✓ Lead	✓ Support
SQL Queries & JOINS	✓ Lead	
ER & UML Diagrams	✓ Lead	✓ Support
Normalization	✓ Lead	
React.js Frontend	✓ Support	✓ Lead
Node.js / Express API		✓ Lead
Frontend-DB Integration	✓ Support	✓ Lead
Sample Data Insertion	✓ Lead	✓ Support
Project Report	✓ DB sections	✓ Frontend sections
Viva Preparation	✓ Both members equally	✓ Both members equally

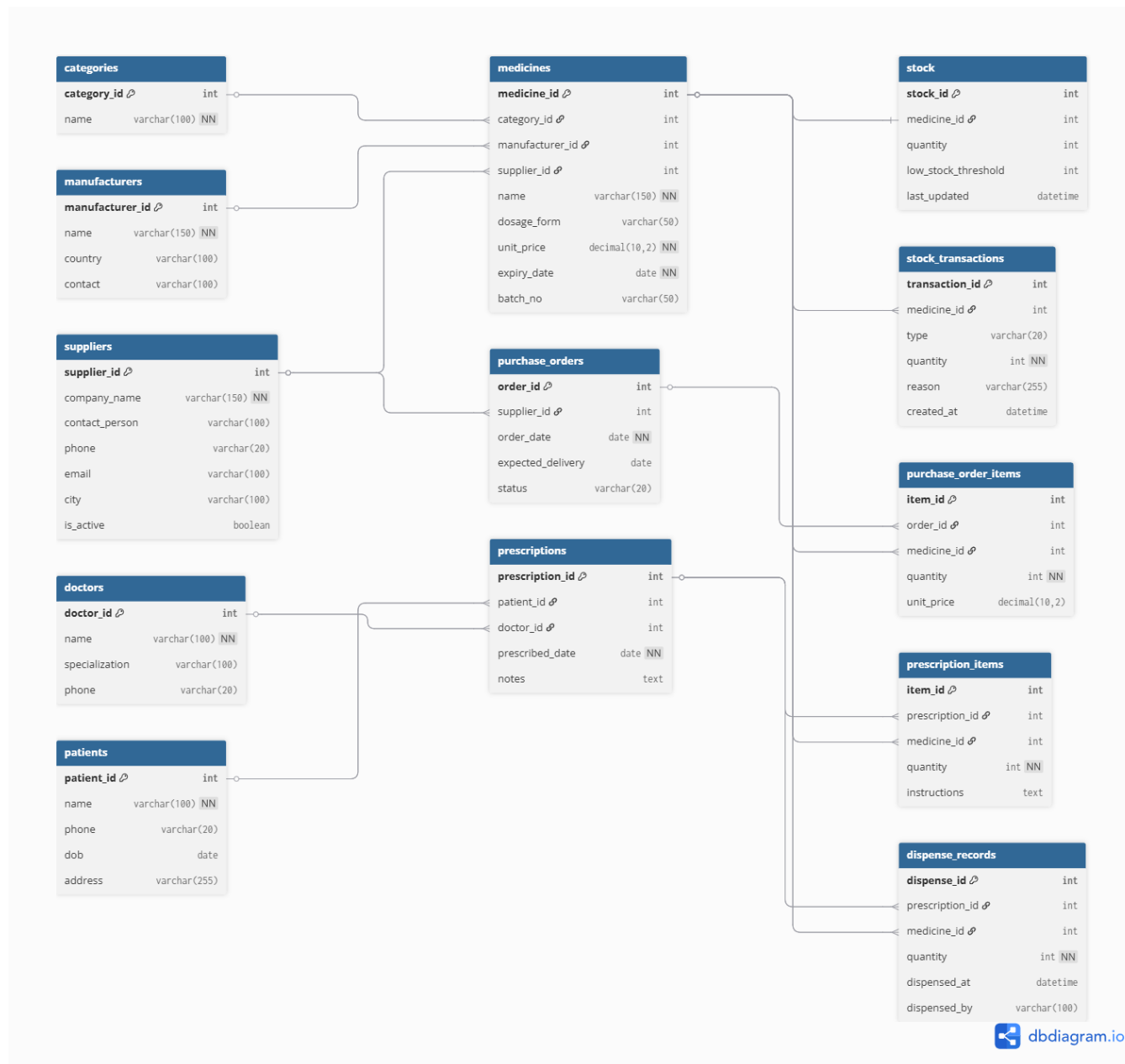
10. Expected Outcomes

Upon completion, this project will deliver:

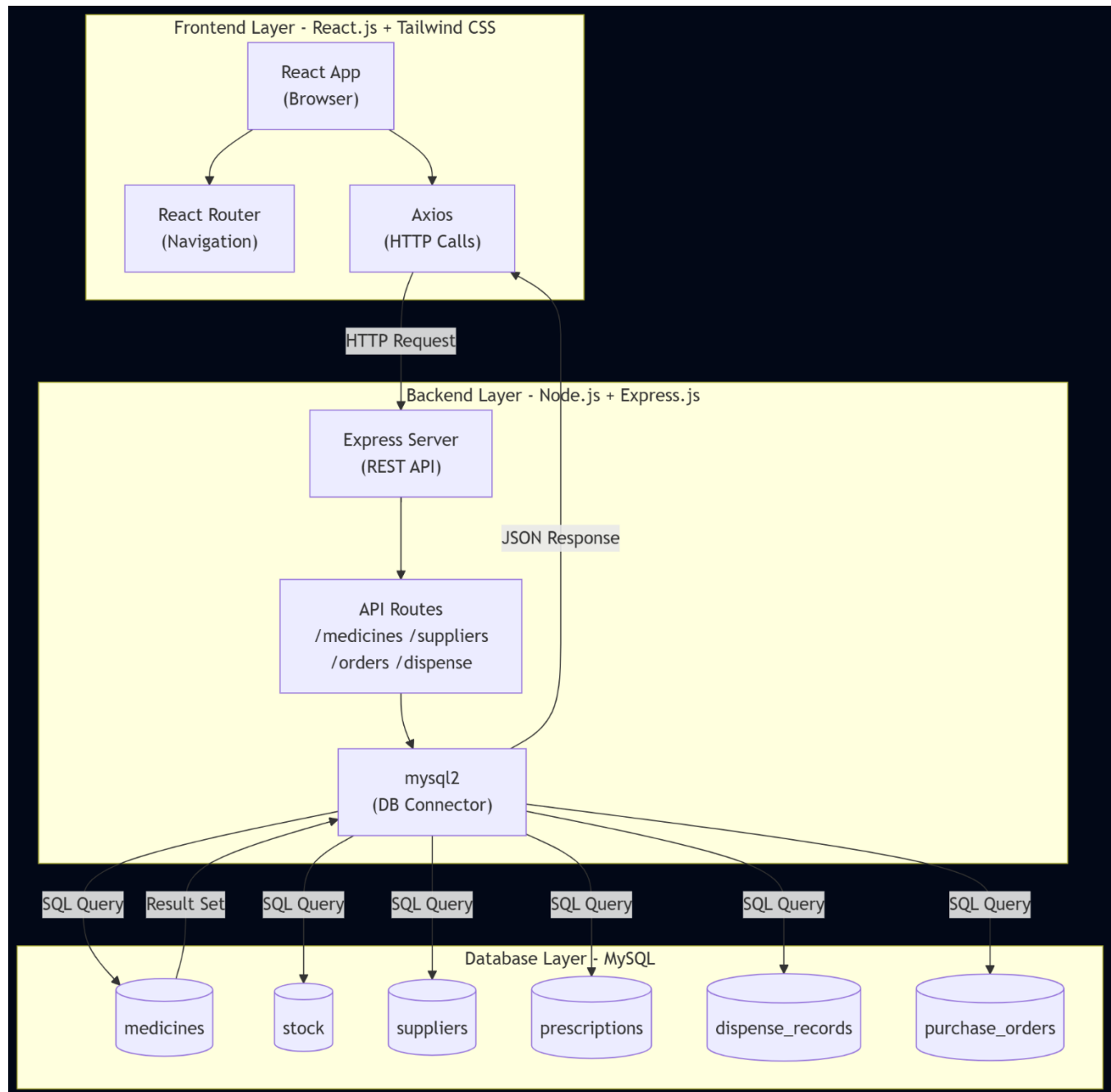
8. A fully functional Hospital Pharmacy & Medicine Inventory Management System with 8 operational modules.
9. A normalized MySQL database with 13 tables, proper constraints, foreign keys, and query-optimized indexes.
10. A minimum of 25 SQL queries demonstrating SELECT, INSERT, UPDATE, DELETE, multi-table JOINS, subqueries, and aggregation.
11. A React.js frontend connected to a Node.js/Express.js backend, demonstrating complete frontend-to-database data integration.
12. An exportable .sql database file demonstrating portability across MySQL-compatible systems.
13. A comprehensive 5-page project report covering database design, query optimization, normalization justification, UI design, and sample query results.
14. A 10–15 minute live viva demonstration showing real-time query execution, data insertion, retrieval, and manipulation.

This project directly addresses a genuine real-world problem in healthcare infrastructure and provides practical hands-on experience with relational database design, SQL query writing, and full-stack web development — skills that are directly applicable in the industry.

ERD DIAGRAM



SYSTEM ARCHITECTURE



Signatures

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